

Definition: The sets A and B are *disjoint* (mutually exclusive) if they have no common elements, hence their intersection is empty: $A \cap B = \emptyset$

Definition: Unions and intersections are *associative*:

$$A \cup B \cup C = A \cup (B \cup C) = (A \cup B) \cup C$$

$$A \cap B \cap C = A \cap (B \cap C) = (A \cap B) \cap C$$

Definition: Intersections and unions follow the distributive laws:

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

Definition: The *universal set*, often denoted by S , is the set that contains all members of a number of related sets under consideration. The exact definition of the universal set depends on the context.

$$A \cap S = A \text{ and } A \cup S = S$$

Definition: Let S represent a universal set. If $A \subseteq S$, then the *complement* of A , denoted by A^c , is the set of all elements of S that are not in A .

$$A^c = \{x : x \notin A\}$$

$$A \cap A^c = \emptyset \text{ and } A \cup A^c = S$$

$$(A^c)^c = A$$

De Morgan's Laws

Theorem: Let A and B be sets. Then De Morgan's Laws state that

$$(A \cup B)^c = A^c \cap B^c$$

$$(A \cap B)^c = A^c \cup B^c$$

Further generalising the theorem, $\left(\bigcup_{n=1}^k A_n\right)^c = \bigcap_{n=1}^k (A_n)^c$ and $\left(\bigcap_{n=1}^k A_n\right)^c = \bigcup_{n=1}^k (A_n)^c$

Partitions

Definition: A collection of sets, $A_1, \dots, A_k$ form a partition of a universal set S if their union is equal to S

$$A_1 \cup A_2 \dots \cup A_k = S$$

and if they are mutually disjoint

$$A_i \cap A_j = \emptyset \quad \forall i \neq j$$

Cardinality

Theorem: The inclusion-exclusion principle (IEP) states that if A and B are sets

of finite cardinality, then

$$|A \cup B| = |A| + |B| - |A \cap B|$$

Theorem: The three-set IEP is such that

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

Cartesian Products and the Product Principle

Definition: In an ordered pair, denoted (a, b) , we have $(a, b) \neq (b, a)$ if $a \neq b$

Definition: The Cartesian product of two sets A and B , denoted $A \times B$ is given by

the set

$$A \times B = \{(a, b) : a \in A \text{ and } b \in B\}$$

The definition can be extended to

$$A_1 \times A_2 \times \dots \times A_n = \{(a_1, a_2, \dots, a_n) : a_1 \in A_1, a_2 \in A_2, \dots, a_n \in A_n\}$$

Definition: The product principle of counting states that when $A_1, A_2, \dots, A_n$ are

finite sets, then the cardinality of their Cartesian product is equal to the product of their

individual cardinalities:

$$|A_1 \times A_2 \times \dots \times A_n| = |A_1| \times |A_2| \times \dots \times |A_n|$$

Permutations and combinations

Definitions: Denote by B the set of balls $\{b_1, b_2, \dots, b_n\}$ with cardinality $|B| = n$. To

draw k balls using the *ordered sampling with replacement* procedure: Draw a ball from

the urn, record which ball was drawn, and then return the ball back to the urn, repeating this

process until we have drawn k balls.

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draw k balls using the *ordered sampling without replacement* procedure: Draw a ball

from the urn and record which ball was drawn, repeating this process until we have drawn k

balls. We do not return a ball once it has been drawn from the urn.

Definition: The ordered sequence of k elements, selected without replacement, from a

set with n elements is called a *permutation*. The number of possible permutations is given by:

$$M(n, k) = \prod_{i=0}^{k-1} (n-i) = \frac{n!}{(n-k)!}$$

Definition: Consider a container with n balls. To draw k balls using the *unordered*

sampling without replacement procedure: Draw k balls from the container

without returning a ball to the container once drawn.

Definition: The unordered sequence of k elements, selected without replacement, from

a set with n elements is called a *combination*. The number of possible combinations, $C(n, k)$

is given by:

$$C(n, k) = \binom{n}{k} = \frac{n!}{(n-k)! k!}$$

Theorem: Binomial Theorem

$$(x+y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k}$$